Prana Earth

Climate Risk Assessment Methodology

Physical Climate Risk Scoring Framework — v2.1

June 2026 | Confidential & Proprietary
For: Prana Earth Platform Development

1. Executive Summary

This document defines the complete methodology for calculating physical climate risk scores for Indian cities, districts, and industrial sites across six climate hazard types. Each hazard is scored on a 1–100 scale using a multi-indicator weighted composite derived from CMIP6 global climate model outputs, India-specific observational databases, and remote sensing data layers.

The framework is designed to serve Prana Earth's primary use case: water resilience and physical climate risk intelligence for data center operators and industrial water users in India. Scores are produced for five SSP scenarios (SSP1-1.9 through SSP5-8.5) at three-time horizons (2030, 2040, 2050) at a minimum 1×1-degree spatial resolution, with city-level refinement to 0.1°.

Six Hazard Domains

Flood Risk — Fluvial, pluvial, and coastal inundation exposure

Heat Stress — Physiological and infrastructural thermal load

Water Stress — Demand-supply imbalance across surface and groundwater

Drought Risk — Meteorological, agricultural, and hydrological drought

Sandstorm / Thunderstorm Risk — Convective storm and dust aerosol hazard

Wildfire Risk — Ignition probability, fuel availability, and fire spread potential

2. Framework Architecture

The scoring pipeline follows a four-stage architecture: raw climate data extraction, indicator computation, hazard score aggregation, and final risk score calculation incorporating exposure and vulnerability overlays.

2.1 Data Pipeline Overview

1. CMIP6 ensemble extraction — pull bias-corrected model outputs for all five SSP scenarios at daily/monthly resolution.
2. Indicator derivation — compute derived indicators (e.g. HWD, SPI, CAPE, FWI) from raw climate variables using domain-standard formulas.
3. Normalisation — min-max scale all indicators to [0, 100] against the India-wide observed baseline (1985–2014).
4. Weighted composite — multiply each normalised indicator by its expert-calibrated weight; sum to produce a raw Hazard score.
5. Non-linear adjustment — apply mild convex transformation to prevent score compression at high values.
6. Final risk score — overlay Hazard score with Exposure (population/asset density) and Adaptive Capacity adjustments.

2.2 CMIP6 Model Selection

Use a minimum of 3 CMIP6 General Circulation Models (GCMs) with demonstrated skill over the Indian subcontinent and South Asian monsoon system. Recommended ensemble:

- MRI-ESM2-0 (Japan Meteorological Research Institute) — Best Indian Ocean SST and monsoon circulation skill
- EC-Earth3 (European consortium) — Strong ENSO-monsoon teleconnection representation
- GFDL-ESM4 (NOAA, USA) — High ocean-atmosphere coupling fidelity
- MPI-ESM1-2-HR (Max Planck Institute) — High horizontal resolution (~0.9°)
- IPSL-CM6A-LR (Institut Pierre-Simon Laplace) — Good aerosol forcing representation

Compute the ensemble median (not mean) for all derived indicators. The median is less sensitive to outlier models and avoids artificial smoothing of extremes that the mean introduces.

2.3 Bias Correction Protocol

Apply Quantile Delta Mapping (QDM) following the ISIMIP3b protocol. Bias-correct all temperature and precipitation variables against the IMD Gridded Observations (0.25° resolution, 1901–present) as the observational baseline.

QDM formula:

```
x_bc = x_obs,hist + F_obs,hist^(-1)( F_GCM,hist(x_GCM) ) × (x_GCM,fut /
x_GCM,hist)
```

Regrid all CMIP6 outputs to a common 0.1° grid using bilinear interpolation for spatial consistency across risk layers. For rapid prototyping, the ISIMIP3b pre-bias-corrected dataset (available at isimip.org) provides ready-to-use outputs at 0.5° for all five SSPs.

2.4 Time Horizons

Compute all indicators as 20-year rolling averages centred on the following epochs:

- Near-term: 2020–2039 (centred on 2030)
- Mid-century: 2030–2049 (centred on 2040)
- Long-term: 2040–2059 (centred on 2050)

Rolling averages suppress inter-annual variability and align with typical asset lifecycle planning horizons for data centers (10–20-year capital commitment windows).

3. SSP Scenario Coverage

Risk scores are generated for all five Shared Socioeconomic Pathway (SSP) scenarios from CMIP6, enabling users to compare outcomes under a range of future emissions trajectories.

Scenario	Warming by 2100	Narrative	Recommended Use
SSP1-1.9	~1.5°C	Sustainability & rapid decarbonisation	Optimistic policy target scenario
SSP1-2.6	~2.0°C	Low forcing, strong mitigation	Paris Agreement aligned baseline
SSP2-4.5	~2.7°C	Middle of the road	Recommended base case for planning
SSP3-7.0	~3.6°C	High forcing, fragmented world	Current policy trajectory stress test
SSP5-8.5	~4.4°C	Fossil-fuelled development	Worst-case scenario; max anchor for normalisation

Dashboard presentation guidance: SSP2-4.5 as 'Base Case', SSP1-2.6 as 'Optimistic Policy', and SSP5-8.5 as 'Stress Test'. SSP3-7.0 best represents the current global policy trajectory and should be surfaced prominently for enterprise clients making 15–20 year capex decisions.

4. Flood Risk

HAZARD TYPE: FLUVIAL · PLUVIAL · COASTAL INUNDATION

Flood risk captures exposure to three distinct flood mechanisms: fluvial flooding driven by river overflow, pluvial flooding from extreme precipitation overwhelming surface drainage, and coastal inundation from sea-level rise and storm surge. India's monsoon regime, flat riverine plains, and rapidly urbanising cities make flood risk the single highest-impact physical hazard for both populations and infrastructure.

4.1 Key Indicators (5 factors)

Factor	Description & Computation	CMIP6 / other Variable(s)	Weight
rx5day — Max 5-day precipitation	Maximum 5-day accumulated rainfall per year (mm). Primary driver of fluvial and pluvial flooding. Captures the intensity of monsoon pulses that overwhelm river capacity.	`pr → rolling 5-day sum → annual max`	**35%**
pr99p — Extreme daily intensity	Daily rainfall above the 99th percentile (mm/day). Captures sudden cloudbursts characteristic of Indian monsoon. Relevant for urban pluvial flash floods.	`pr → JJA* percentile per grid`	**25%**
DEM Slope + TWI — Topographic susceptibility	Slope (degrees) and Topographic Wetness Index from 30m SRTM DEM. Low slope + high TWI = high ponding risk. Static layer resampled to 0.1°.	`SRTM/ALOS DEM (static)`	**20%**
mrso — Antecedent soil moisture	Mean soil moisture (kg/m²) in top 30 cm during pre-monsoon. Saturated soils amplify runoff from subsequent rainfall events.	`mrso (top layer) · ISIMIP3b bias-corrected`	**12%**
Water Logging	Soil salinity preventing percolation of water in the water table	`??`	**??**
NDVI + Impervious % — Drainage capacity	Urban impervious fraction reduces infiltration; NDVI captures natural drainage function. Combined metric: (1 - NDVI) × impervious_fraction.	`ESA CCI Land Cover + Landsat NDVI`	**8%**

*JJA: Jun, Jul, Aug

4.2 CMIP6 Variables Required

- pr — Precipitation flux (kg/m²/s) · daily resolution
- mrso — Total soil moisture content · monthly
- mrros — Surface runoff · monthly
- psl — Sea level pressure (for cyclonic surge modelling)

4.3 India-Specific Data Sources

- NRSC BHUVAN — National floodplain delineation maps at 1:50,000 scale for major river basins (bhuvan.nrsc.gov.in)
- CWC HIS — Central Water Commission historical river gauge discharge for return period analysis (cwc.gov.in)
- IMD Gridded Rainfall (0.25°) — Observed 1901–present daily rainfall for bias correction baseline

5. Heat Stress

HAZARD TYPE: PHYSIOLOGICAL + INFRASTRUCTURAL THERMAL LOAD

Heat stress captures both the direct physiological risk to outdoor workers and urban populations, and the indirect infrastructural risk from elevated temperatures — cooling energy demand, transformer failures, and water-cooling system constraints for data centers. India's pre-monsoon heat wave season and intensifying urban heat islands make this a rapidly growing risk class.

5.1 Key Indicators (5 factors)

Factor	Description & Computation	CMIP6 / other Variable(s)	Weight
HWD — Heat wave duration	Annual count of days in heat wave events (Tmax ≥ 4.5°C to 6.4°C above normal for ≥3 consecutive days, following IMD definition). Best single proxy for human heat mortality risk.	`tasmax → IMD heat wave threshold (≥4.5°C above normal)`	**30%**
WBT/WBGT — Wet bulb temperature	Wet Bulb Globe Temperature (°C). Integrates heat and humidity effects on thermoregulation. Critical for outdoor labour exposure. At WBGT > 35°C, human thermoregulation fails entirely.	`tas + hurs → Stull (2011) WBT approximation`	**30%**
TXx — Annual peak temperature	Annual maximum of daily maximum temperature (°C). Determines peak thermal stress on infrastructure: cooling system sizing, transformer load limits, data center PUE degradation.	`tasmax → annual maximum`	**20%**
CDD — Cooling degree days	Sum of (Tmean − 18°C) on days Tmean > 18°C. Directly drives energy demand for space cooling. Proxy for data center cooling system workload and PUE deterioration under climate change.	`tas → cumulative sum above 18°C base`	**12%**
UHI proxy — Night temperature anomaly	Mean minimum temperature (tasmin) relative to rural surroundings using GHSL urban density overlay. Urban areas retain heat nocturnally — UHI amplifies heat risk in data center zones by 2–5°C.	`tasmin + GHSL urban density → urban-rural delta T`	**8%**

5.2 CMIP6 Variables Required

- tasmax — Daily maximum near-surface air temperature (K)
- tasmin — Daily minimum near-surface air temperature (K)
- tas — Daily mean near-surface air temperature (K)
- hurs — Near-surface relative humidity (%)
- huss — Near-surface specific humidity (kg/kg) — for WBGT computation

5.3 India-Specific Data Sources

- IMD Heatwave Advisories + 700+ AWS Station Data — Official heat wave declarations; bias correction baseline (imdpune.gov.in)
- GHSL — Global Human Settlement Layer — Urban density grids for UHI multiplier at 100m resolution (ghsl.jrc.ec.europa.eu)
- MODIS LST (Terra/Aqua MOD11A1) — 1km land surface temperature for UHI validation and real-time heat index overlay

6. Water Stress

HAZARD TYPE: DEMAND–SUPPLY IMBALANCE · SURFACE + GROUNDWATER

Water stress measures the imbalance between total water demand (agricultural, industrial, municipal) and available renewable water supply from both surface water bodies and groundwater aquifers. India faces an acute structural water crisis: it hosts 18% of the world's population on 4% of global freshwater. The Indo-Gangetic Plain groundwater aquifer is depleting at an alarming rate, making India WRIS groundwater monitoring data a critical competitive differentiator for Prana Earth.

6.1 Key Indicators (5 factors)

Factor	Description & Computation	CMIP6 / other Variable(s)	Weight
BWS — Baseline Water Stress	Ratio of total annual water withdrawals to available renewable surface water (WRI Aqueduct 4.0 scale, 0–5+). Ground truth for current physical scarcity. Apply CMIP6 pr/evspsbl delta to project forward.	`WRI Aqueduct 4.0 + CMIP6 pr/evspsbl delta`	**30%**
GWD — Groundwater depth anomaly	Change in groundwater table depth (metres) vs 2010 baseline from India WRIS seasonal monitoring. India uses 89% of extracted groundwater for irrigation — groundwater depletion is the primary long-run water risk driver.	`India WRIS + GRACE-FO TWSA mascon solution`	**25%**
mrro delta — Runoff anomaly	Projected change in annual total runoff (%) vs 1985–2014 baseline. Captures reduction in river flows feeding reservoirs and recharging aquifers. CMIP6 projects 10–30% runoff decline across peninsular India by 2050.	`mrro (surface runoff) · annual mean delta`	**20%**
evspsbl — Evapotranspiration demand	Total evapotranspiration (mm/yr). Rising ET under warming increases consumptive demand, reducing effective water supply even if precipitation is unchanged.	`evspsbl (evaporation + transpiration) · annual sum`	**15%**
pr CV — Monsoon variability	Coefficient of variation of JJAS* precipitation across 20-year windows. High CV = unreliable monsoon = storage planning risk. Linked to Indian Ocean Dipole (IOD) and ENSO teleconnection changes.	`pr (JJAS months only) → CV across 20-yr rolling windows`	**10%**

*JJAS = June, July, Aug, Sept

6.2 CMIP6 Variables Required

- pr — Precipitation flux · daily to monthly aggregation
- evspsbl — Total evapotranspiration (kg/m²/s)
- mrro — Total runoff (kg/m²/s)
- mrsos — Moisture in upper soil layer

6.3 India-Specific Data Sources

- India WRIS — indiawris.gov.in — Groundwater levels (seasonal monitoring), reservoir storage, river discharge, basin water balance. PRIMARY source for groundwater risk.
- GRACE-FO — NASA/DLR mascon solution — Monthly terrestrial water storage anomaly (TWSA) at ~300km. Critical for groundwater depletion quantification in the IGP belt.
- WRI Aqueduct 4.0 — aqueduct.wri.org — CMIP6-driven future water risk; use as cross-validation layer (too coarse for city-level as primary).
- CWC Reservoir Levels (weekly) — Live storage data for 150+ major reservoirs; supply-side validation for seasonal risk updates.

7. Drought Risk

HAZARD TYPE: METEOROLOGICAL · AGRICULTURAL · HYDROLOGICAL

Drought risk encompasses three interrelated drought types. Meteorological drought (precipitation deficit) propagates into agricultural drought (soil moisture deficit causing crop failure) and hydrological drought (reservoir and aquifer depletion). The SPEI index is preferred over SPI for future projections because it captures the warming-amplified evaporative demand that will increasingly drive drought under climate change even without precipitation changes.

7.1 Key Indicators (5 factors)

Factor	Description & Computation	CMIP6 / other Variable(s)	Weight
SPI-12 — Standardised Precipitation Index	12-month SPI: z-score of cumulative precipitation against long-term gamma distribution. SPI ≤ −1.5 = severe drought; ≤ −2.0 = extreme drought. Universal meteorological drought standard.	`pr (monthly) → 12-month rolling sum → gamma fit → SPI`	**30%**
SPEI — Standardised P-ET Index	SPI extended with potential evapotranspiration deficit (using Penman-Monteith PET). Captures warming-amplified drought under climate change better than SPI. The preferred index for future scenarios.	`pr + tas + hurs + rsds → PET (PM method) → P-PET balance → SPEI`	**25%**
mrso anomaly — Root zone soil moisture deficit	Root zone soil moisture below 20th percentile of baseline distribution. Direct indicator of agricultural drought and crop failure risk. Most actionable indicator for irrigation demand forecasting.	`mrso (all soil layers summed) → percentile anomaly vs baseline`	**25%**
CDD — Consecutive dry days	Maximum number of consecutive days with precipitation < 1 mm/day per year. Measures drought persistence and duration; high CDD in Kharif season (Jun-Sep) = direct crop failure risk.	`pr → daily threshold < 1mm → max run-length per year`	**12%**
pr trend — Long-term precipitation trend	Sen's slope of annual precipitation (mm/decade) tested with Mann-Kendall. Declining trend identifies regions undergoing chronic drying and potential reclassification toward arid climate regimes.	`pr (annual) → Mann-Kendall trend + Sen slope per grid cell`	**8%**

7.2 CMIP6 Variables Required

- pr — Precipitation flux · monthly aggregation
- tas — Mean surface air temperature
- mrso — Soil moisture content (all layers)
- evspsbl — Evapotranspiration (for PET computation)
- rsds — Surface downwelling shortwave radiation (for Penman-Monteith)

7.3 India-Specific Data Sources

- IMD Drought Monitoring — Monthly district-level SPI maps and official drought declarations; validation ground truth
- NDMC — National Drought Monitoring Centre — Soil moisture anomaly maps and crop stress indicators

8. Sandstorm / Thunderstorm Risk

HAZARD TYPE: CONVECTIVE STORM + DUST AEROSOL MOBILISATION

This composite risk type covers two distinct but climatologically related hazards: (1) convective thunderstorms driven by atmospheric instability (CAPE), characteristic of India's pre-monsoon season across the northern plains; and (2) dust/sandstorm events driven by bare soil exposure and high wind speeds, concentrated in Rajasthan, Gujarat, and Haryana. Both hazards are amplified by climate change through increased moisture-heat gradients and expanding arid zones respectively.

8.1 Key Indicators (5 factors)

Factor	Description & Computation	CMIP6 / other Variable(s)	Weight
CAPE — Convective Available Potential Energy	Thermodynamic energy available for convective storm development (J/kg). CAPE > 1,500 J/kg = high severe thunderstorm risk; > 2,500 J/kg = extreme. Computed from vertical atmospheric profiles.	`ta + hus at multiple pressure levels → parcel method CAPE. ERA5 reanalysis for validation.`	**30%**
pr99p — Extreme precipitation intensity	99th percentile daily rainfall. Proxies thunderstorm and cloudburst frequency. Particularly relevant in pre-monsoon (Mar-May) across NW India and for the eastern monsoon onset period.	`pr (daily) → seasonal percentile (MAM, JJA separately)`	**20%**
sfcWind p90 — Wind speed extremes	90th percentile of daily mean 10m wind speed (m/s). High wind + dry soil = dust mobilisation. High wind + high CAPE = storm damage amplifier for structures and power infrastructure.	`sfcWind → seasonal 90th percentile per grid cell`	**20%**
Dust emission proxy — soil erodibility index	Combined metric: (1 − NDVI) × wind_p90 × (1 − mrso_normalised). Captures dust mobilisation potential across exposed bare soils. Critical for Rajasthan-Gujarat-Haryana dust corridors.	`mrso + sfcWind (CMIP6); HWSD soil texture + MODIS NDVI (static)`	**18%**
NDVI trend — Vegetation cover change	Declining NDVI trend (greening-to-browning) identifies expanding bare soil exposure. Strong predictor of shifting dust storm corridors and increasing dust event frequency under climate change.	`MODIS MOD13A3 NDVI (monthly 1km) → Sen slope 2001-present`	**12%**

8.2 CMIP6 Variables Required

- ta (plev) — Air temperature at multiple pressure levels · 6-hourly resolution
- hus (plev) — Specific humidity at pressure levels (for CAPE parcel computation)
- sfcWind — Near-surface wind speed (m/s) · daily
- pr — Precipitation flux · daily

- mrso — Soil moisture (for dust emission suppression factor)

8.3 India-Specific Data Sources

- SAFAR (System of Air Quality and Weather Forecasting and Research) — Dust AOD and storm event records for Delhi, Mumbai, Pune, Ahmedabad (safar.tropmet.res.in)
- IMD Thunderstorm Atlas + Lightning Strike Database — Pre-monsoon convective storm frequency maps by district

9. Wildfire Risk

HAZARD TYPE: IGNITION PROBABILITY · FUEL AVAILABILITY · FIRE SPREAD

Wildfire risk in India is concentrated in deciduous and semi-evergreen forest zones of central and peninsular India, with a distinct fire season from January to May in peninsular India and October to November in the northeast. Climate change is extending fire seasons, increasing fire weather frequency, and expanding the geographic footprint of high fire risk zones. This has significant implications for data center operators near forested catchments that depend on those forests for water supply.

9.1 Key Indicators (5 factors)

Factor	Description & Computation	CMIP6 Variable(s)	Weight
FWI — Fire Weather Index (Canadian)	Composite fire danger rating integrating temperature, humidity, wind, and precipitation. International standard (used by FAO and GFMC). Computed using daily noon-time meteorological conditions.	`tasmax + hurs + sfcWind + pr → FWI daily computation`	**30%**
VPD — Vapour Pressure Deficit	Difference between saturation and actual vapour pressure (kPa). High VPD desiccates fuel and amplifies fire spread rate. Strong predictor of fire severity and area burned across Indian forest types.	`tas + hurs → VPD = 0.6108 × exp(17.27T/(T+237.3)) × (1 - RH/100)`	**25%**
FFDI — Forest Fire Danger Index (McArthur)	Australian-derived index validated for South Asian tropical forests. Integrates drought factor (D), temperature, humidity, and wind. Formula: 2 × exp(-0.45 + 0.987×ln(D) - 0.0345×H + 0.0338×T + 0.0234×V).	`tasmax + hurs + sfcWind + mrso (for drought factor D)`	**20%**
LFMC — Live fuel moisture content	Live fuel moisture (%) derived from MODIS NIR/SWIR band ratio. LFMC < 80% = ignition-ready fuel. Combined with NDVI biomass proxy to estimate total fuel load per grid cell.	`MODIS MOD09A1 surface reflectance → LFMC model; mrso as climate proxy`	**15%**
sfcWind p90 — Fire-season wind speed	90th percentile daily wind speed during India fire season (Jan-May, Oct-Nov peninsular). Wind is the primary driver of fire spread rate, spotting distance, and suppression difficulty.	`sfcWind → p90 restricted to fire season months`	**10%**

9.2 CMIP6 Variables Required

- tasmax — Daily maximum temperature (for FWI and FFDI)
- hurs — Near-surface relative humidity · daily
- sfcWind — Wind speed at 10m · daily
- pr — Precipitation (for drought factor D computation)
- mrso — Soil moisture (drought factor D proxy)

9.3 India-Specific Data Sources

- FSI (Forest Survey of India) — Fire alert system; historical fire point locations for threshold calibration (fsi.nic.in)
- NASA FIRMS VIIRS I-Band — 375m near-real-time active fire detections; build fire climatology per city for wildfire validation (firms.modaps.eosdis.nasa.gov)
- ISFR (India State of Forest Report) — Biennial district-level forest cover and forest type maps for fuel classification input

10. Scoring Methodology

This section defines the complete quantitative pipeline from raw CMIP6 outputs to the final 1–100 risk score for each hazard type.

10.1 Step 1 — Normalisation

All computed indicators are normalised to a common [0, 100] scale using min-max normalisation anchored to the India-wide observed distribution.

Normalisation formula:

```
I_norm = 100 × (I_raw - I_min) / (I_max - I_min)
```

- I_min = 5th percentile of indicator across all Indian grid cells during 1985–2014 baseline
- I_max = 95th percentile of SSP5-8.5 projections for 2050 (hard cap to prevent unbounded scores)
- Indicators where lower values mean higher risk (e.g. NDVI, mrso) are inverted before normalisation

10.2 Step 2 — Weighted Composite (Hazard Score)

The normalised indicators are combined into a single Hazard Score using expert-calibrated weights validated against EM-DAT historical disaster loss data and NDMA records.

```
Hazard_score = sum_i (w_i × I_norm_i)    where sum_i(w_i) = 1.0
```

Weights for each risk type are defined in Sections 4–9 respectively.

10.3 Step 3 — Non-Linear Adjustment

A mild convex transformation prevents score compression at high values, ensuring cities at the 90th percentile of risk genuinely score near 90 rather than clustering in a narrow range around 60–70.

```
H_adj = 100 × (H_raw / 100) ^ 0.85
```

The exponent 0.85 is a calibration parameter. Recalibrate using local validation data if available (e.g. compare H_adj flood scores against observed flood frequency from CWC).

10.4 Step 4 — Final Risk Score (Hazard × Exposure × Vulnerability)

The final risk score incorporates three components following the IPCC risk framework: Hazard (physical climate signal), Exposure (people and assets in harm's way), and inverse Adaptive Capacity (ability to cope and recover).

```
Risk_final = 0.60 × H_adj + 0.20 × Financial_Exposure_norm + 0.10 ×
Population_Exposure_norm - 0.10 × AdaptCap_norm
```

- Financial_Exposure layer: (Asset value at location / National avg asset value) × 100
- Population_Exposure layer: WorldPop 100m population density (normalised) OR Floor Area Ratio (FAR) as asset density proxy for data center applications
- Adaptive Capacity: composite of NDVI (green cover fraction) + census income quintile rank (normalised to [0,1])
- The 0.70 / 0.20 / 0.10 weights are initial assumptions. Recalibrate using regression against observed disaster loss per unit exposure.

10.5 Output Format

Each scored location produces a JSON record:


```
{ "city": "Hyderabad", "lat": 17.38, "lon": 78.49,
  "scenario": "SSP2-4.5", "horizon": 2040,
  "flood": 67, "heat_stress": 84, "water_stress": 72,
  "drought": 58, "storm": 41, "wildfire": 29,
  "composite_risk": 64 }
```

11. Complete Data Source Register

11.1 Primary — CMIP6 Climate Model Outputs

Source	URL / Access	Notes
ISIMIP3b	isimip.org isimip.org/outputdata/isimip-repository/	Bias-corrected CMIP6 at 0.5°, all 5 SSPs. Recommended for hydrological vars (mrso, evspsbl, mrro).
ESGF Node (primary)	esgf-node.llnl.gov / aims3.llnl.gov	Full CMIP6 archive. Use Intake-ESM or ESGF search API. All SSPs available.
Google Earth Engine	NASA/GDDP-CMIP6 collection	Pre-processed monthly tas, tasmax, pr at 0.25°. Fastest path for 1° analysis.

11.2 India-Specific Observation & Monitoring

Source	URL	Provides
IMD Gridded Obs	imdpune.gov.in	0.25° daily rainfall & temperature, 1901-present. Bias correction baseline.
India WRIS	indiawris.gov.in	Groundwater levels, reservoir storage, river discharge, basin water balance.
CWC HIS	cwc.gov.in	River gauge discharge, flood forecasting, return period analysis.
NRSC Bhuvan	bhuvan.nrsc.gov.in	Floodplain maps, LULC 1:50,000, vegetation, disaster archive.
FSI Fire Alert	fsi.nic.in	Historical fire points, forest cover, forest type classification.
SAFAR	safar.tropmet.res.in	Dust AOD, storm records: Delhi, Mumbai, Pune, Ahmedabad.
IMD Thunderstorm Atlas	imd.gov.in	Pre-monsoon convective storm frequency by district.
NDMC (Drought)	imd.gov.in/ndmc	Soil moisture anomaly maps, official drought declarations.

11.3 Global Satellite & Reanalysis

Source	Access	Provides
ERA5 (ECMWF)	cds.climate.copernicus.eu	Hourly/monthly reanalysis at 0.25°. CAPE, wind extremes, bias cross-check.
GRACE-FO (NASA/DLR)	grace.jpl.nasa.gov	Monthly TWSA at 0.5°. Best available groundwater proxy for IGP.
MODIS (Terra/Aqua)	earthdata.nasa.gov	NDVI (MOD13A3), LST (MOD11A1), surface reflectance (MOD09A1).
NASA FIRMS VIIRS	firms.modaps.eosdis.nasa.gov	375m active fire detections near-real-time.
SRTM / ALOS DEM	opentopography.org / GEE	30m elevation; slope, TWI, drainage accumulation.
GHSL (JRC)	ghsl.jrc.ec.europa.eu	Built-up area, building height, urban centre classification.

11.4 Risk Intelligence & Cross-Check Layers

Source	URL	Use Case
WRI Aqueduct 4.0	aqueduct.wri.org	Cross-validation for water stress. Too coarse for primary city-level use.
EM-DAT	emdat.be	Historical disaster losses. Essential for threshold calibration and model validation.
WorldPop	worldpop.org	100m gridded population density. Exposure layer computation.
HWSD Soil Texture	fao.org/soils-portal	Soil erodibility classification for dust emission proxy.

12. Implementation Notes & Recommended Models

12.1 Recommended CMIP6 GCMs for India

Model	Institution	Strength for India
MRI-ESM2-0	JMA, Japan	Best Indian Ocean SST and monsoon circulation skill
EC-Earth3	European consortium	Strong ENSO-monsoon teleconnection representation
GFDL-ESM4	NOAA, USA	High ocean-atmosphere coupling fidelity
MPI-ESM1-2-HR	MPI, Germany	Highest horizontal resolution (~0.9°) in ensemble
IPSL-CM6A-LR	IPSL, France	Good aerosol forcing; useful for dust/storm risk

12.2 Technology Stack Recommendations

- Python (xarray + intake-esm) — CMIP6 data access and manipulation
- Google Earth Engine (Python/JS API) — MODIS, NDVI, LST, and CMIP6 GDDP processing
- SciPy / statsmodels — SPI/SPEI computation, Mann-Kendall trend tests
- ISIMIP3b portal — Pre-bias-corrected CMIP6 download (avoids custom QDM pipeline for v1)
- PostGIS / PostgreSQL — Spatial storage of scored grid cells
- FIRMS API — Real-time fire overlay integration

12.3 Validation Approach

Before production deployment, validate computed risk scores against observed historical event data:

7. Flood: Compare normalised flood scores against CWC discharge exceedance records and NDMA flood-affected district data (2000–2023).
8. Heat: Compare HWD scores against IMD heat wave death toll records by district.
9. Drought: Compare SPI-12 / SPEI scores against NDMC declared drought districts by year.
10. Wildfire: Compare FWI / FFDI scores against FSI fire point density by forest division.
11. Water: Compare BWS + GWD scores against CGWB groundwater over-exploited block classifications.

Use the Spearman rank correlation coefficient (rho) as the primary validation metric. Target rho ≥ 0.65 for each risk type against the corresponding observational record before production release.

12.4 Limitations & Caveats

- CMIP6 models underestimate Indian summer monsoon variability — ensemble spread is large; communicate uncertainty ranges to users, not just point estimates.
- 1×1° resolution is insufficient for micro-scale urban flood modelling — flag cities where hyper-local flood modelling (SWMM, HEC-RAS) would meaningfully improve accuracy.
- Groundwater projections have the highest uncertainty — GRACE-FO TWSA does not distinguish between aquifer layers; field validation through India WRIS monitoring wells is essential.
- Wildfire risk is lowest-confidence for India given sparse historical calibration data compared to Australia or California.
- Exposure and Adaptive Capacity layers use static snapshots — update annually as WorldPop and census data refresh.

Appendix A — CMIP6 Variable Quick Reference

Variable ID	Full Name	Units	Used In
pr	Precipitation flux	kg/m²/s	Flood, Water, Drought, Storm, Wildfire
tas	Mean near-surface air temp	K	Heat, Drought, Water
tasmax	Max near-surface air temp	K	Heat, Wildfire
tasmin	Min near-surface air temp	K	Heat (UHI)
hurs	Near-surface relative humidity	%	Heat, Wildfire, Storm
huss	Near-surface specific humidity	kg/kg	Heat (WBGT)
mrso	Total soil moisture	kg/m²	Flood, Drought, Water, Storm, Wildfire
mrros	Surface runoff	kg/m²/s	Flood
mrro	Total runoff	kg/m²/s	Water
evspsbl	Evapotranspiration	kg/m²/s	Water, Drought
sfcWind	Near-surface wind speed	m/s	Storm, Wildfire
psl	Sea level pressure	Pa	Flood (coastal)
ta (plev)	Air temp at pressure levels	K	Storm (CAPE)
hus (plev)	Specific humidity at pressure levels	kg/kg	Storm (CAPE)
rsds	Surface downwelling SW radiation	W/m²	Drought (PET)

Appendix B — Glossary

Term	Definition
BWS	Baseline Water Stress — total water withdrawals / available renewable supply
CAPE	Convective Available Potential Energy — atmospheric instability metric (J/kg)
CDD (climate)	Cooling Degree Days — cumulative temperature above 18°C base
CDD (drought)	Consecutive Dry Days — max run of days with pr < 1 mm
FFDI	Forest Fire Danger Index (McArthur) — Australian standard for fire danger
FWI	Fire Weather Index (Canadian) — composite fire danger rating system
GWD	Groundwater depth anomaly — change in water table depth vs baseline
HWD	Heat Wave Duration — annual days in qualifying heat wave events

Term	Definition
LFMC	Live Fuel Moisture Content — % moisture in living vegetation
QDM	Quantile Delta Mapping — bias correction methodology (ISIMIP3b)
SPI	Standardised Precipitation Index — z-score of cumulative precipitation
SPEI	SPI extended with evapotranspiration deficit for warming adjustment
SSP	Shared Socioeconomic Pathway — CMIP6 climate forcing scenario
TXx	Annual maximum of daily maximum temperature
TWI	Topographic Wetness Index — terrain-based flood susceptibility metric
TWSA	Terrestrial Water Storage Anomaly (GRACE-FO)
UHI	Urban Heat Island — elevated urban temperatures vs rural surroundings
VPD	Vapour Pressure Deficit — evaporative demand driver (kPa)
WBGT	Wet Bulb Globe Temperature — physiological heat stress index (°C)